

Project 2: Milestone 3

David Koyrakh

Bellevue University

DSC-680: Applied Data Science

Professor Iranitalab

November 2, 2025

Project 2: Milestone 3

Topic

Predicting Home Sale Prices and Discovering Correlations in Ames, Iowa (Ames Housing Dataset).

Business Problem

Accurate home price estimation helps real-estate professionals, appraisers, and homeowners make informed decisions. These estimations inform their decisions around pricing, renovation priorities, and market timing. The objective of this project is to build a transparent and performant model to predict a home's sale price from routine listing and property characteristics, and to communicate which features most strongly drive price. Success will be measured by error on held-out data (RMSE, MAE, R2) and by the clarity of explanations for non-technical stakeholders.

Background and History

The Ames Housing dataset is a widely used benchmark for residential valuation research, compiled by De Cock (2011) and popularized via the Kaggle “House Prices: Advanced Regression Techniques” competition. The data represent 1,460 property sales with 79 explanatory variables and a continuous target, SalePrice, largely spanning the 2006–2010 period in Ames, Iowa. The feature set includes lot and neighborhood descriptors, structure quality and size, basement and garage details, as well as kitchen, bath, and sale conditions. Because the dataset is public, well documented, and diverse in feature types, it is well suited for studying preprocessing choices, linear vs. tree-based modeling, and model explainability.

Data Exploration

The working table has a shape of 1,460 by 81 with the target variable being “SalePrice”. Many predictors are categorical. Several fields encode true absence with “NA” (for example, no alley, no fence, no pool). High-missing columns include “PoolQC” (99.5%), “MiscFeature” (96%), “Alley” (93.8%), and “Fence” (80.8%). Numeric fields such as “LotFrontage” have moderate gaps (17.7%). The target distribution is right-skewed. A log transform yields a near-symmetric shape (mean is approximately 12.02 on log1p scale), which is beneficial for linear models and residual stability.

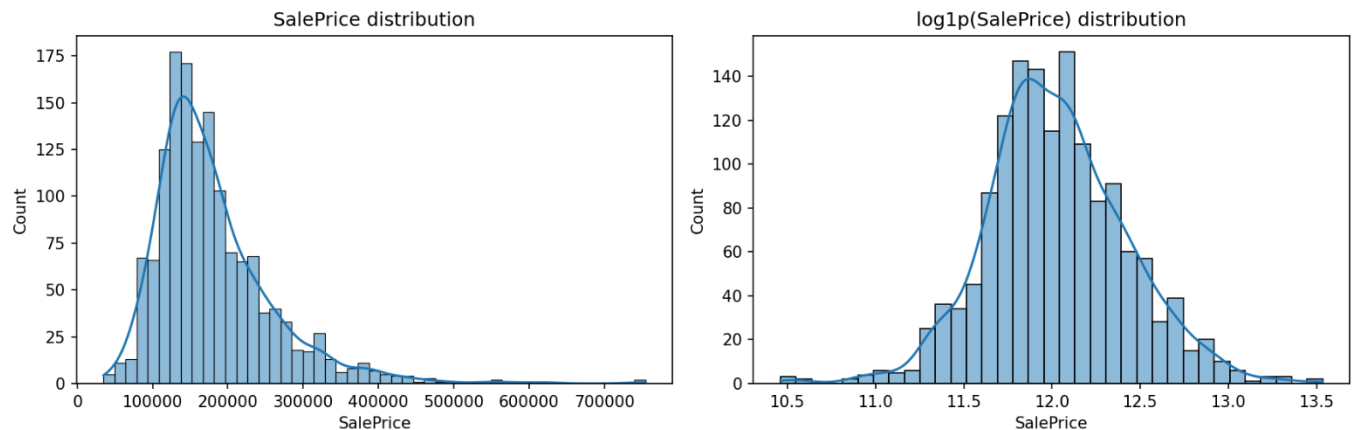


Figure 1. Distribution of SalePrice and log1p(SalePrice)

Research Questions

1. How accurately can I predict a home’s sale price using the provided variables?
2. Which features most strongly correlate with sale price?
3. How do linear models compare to tree-based models?
4. Can I produce stakeholder-friendly explanations to justify my model’s predictions?
5. What are the principal data quality issues and how do they impact the model?

Methods

Data Preparation

The dataset will be split into training, validation, and test sets, using stratified sampling or K-fold cross-validation with a final hold-out set. Missing values will be treated with semantics-aware methods. Explicit “None” categories will be generated for true absences and imputation for numeric gaps. Ordinal quality variables will be encoded consistently with domain meaning. Log transformation of SalePrice and selected skewed predictors will be applied, with additional interaction features and standardization of numeric variables as needed.

Modeling

Baseline models will include the mean predictor and linear regression variants (Ridge, Lasso, Elastic Net) implemented in pipelines. Tree-based approaches such as Random Forest and Gradient Boosting (XGBoost or LightGBM) will follow, with early stopping and hyperparameter tuning. Model selection will rely on cross-validated performance using RMSE as the primary metric, with MAE and R2 as secondary criteria.

Evaluation and Interpretation

Performance will be assessed using RMSE, MAE, and R2, supported by residual diagnostics. Interpretability will draw on standardized coefficients (linear models), permutation importance, partial dependence, ICE plots, and SHAP values for tree-based models. Results will be communicated through clear visuals, and robustness will be tested against outliers, encoding alternatives, and feature subset variations.

Analysis

Exploratory analysis confirms that quality and habitable area strongly co-vary with price, while recency (for example, “YearBuilt”) also plays a meaningful role. Bivariate inspections show visible heteroskedasticity at very large “GrLivArea” values, suggesting potential benefits from robust loss functions or transformation. A correlation heatmap of the top numeric drivers ranks “OverallQual”, “GrLivArea”, floor and basement square footage, and garage capacity near the top, indicating multicollinearity among size measures that regularization can help moderate.

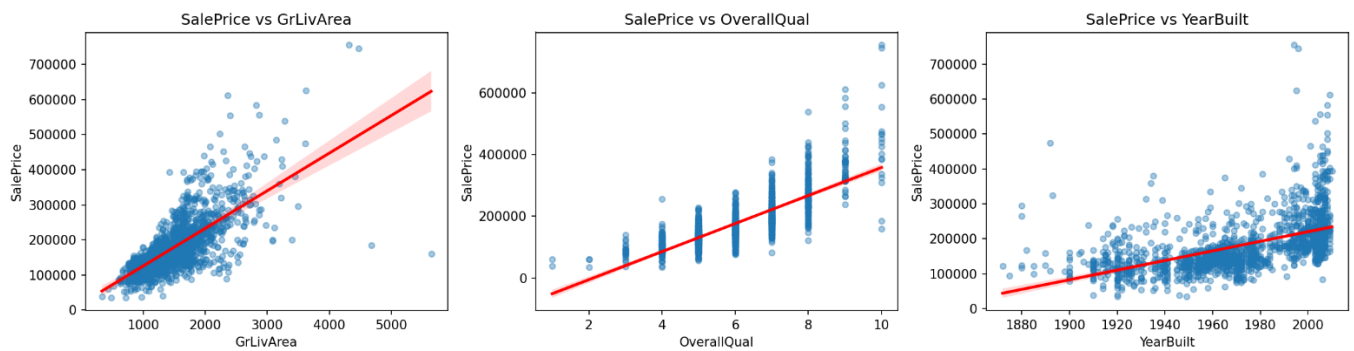


Figure 2. SalePrice vs. key drivers (GrLivArea, OverallQual, YearBuilt).

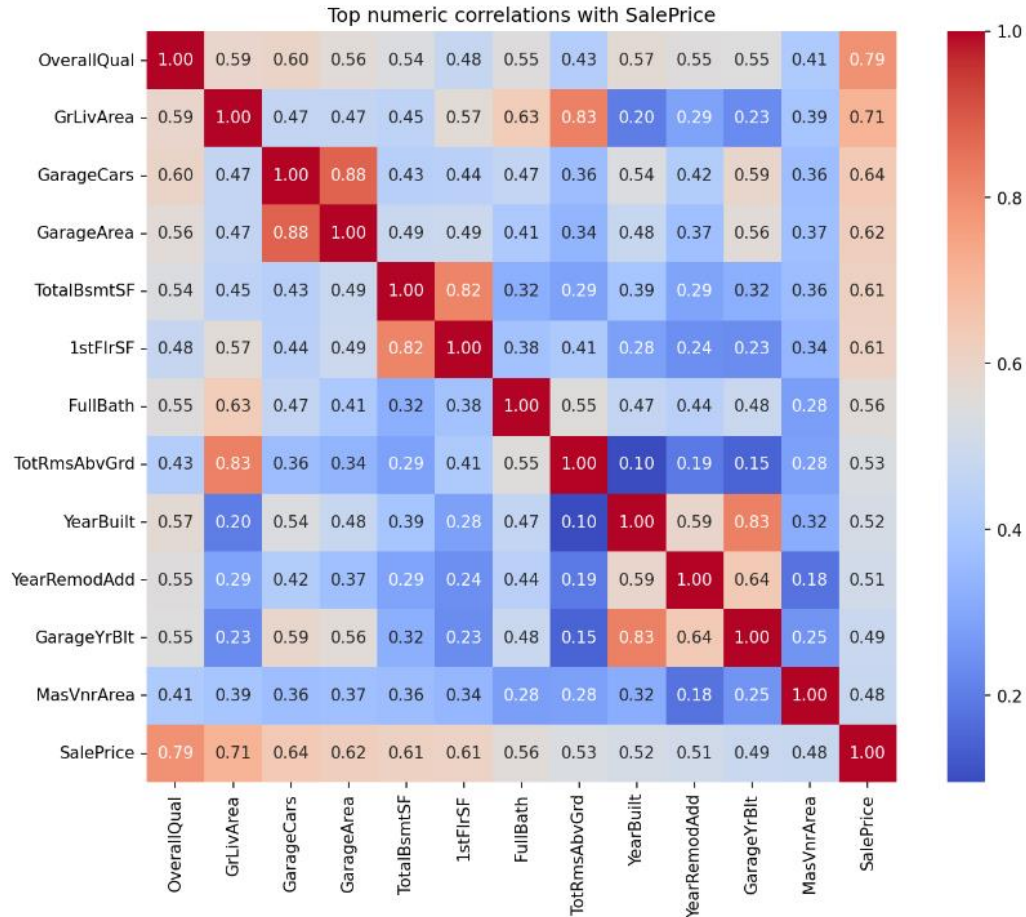


Figure 3. Correlation heatmap of top numeric features with SalePrice.

With straightforward preprocessing (median/mode imputation, scaling, one-hot encoding), the Ridge baseline yields a 5-fold RMSE mean of approximately 33.1K, while the Random Forest baseline achieves approximately 30.2K. The tree model benefits from flexible, non-linear interactions and mixed data types without additional feature engineering. Feature importance from the fitted Random Forest emphasizes “OverallQual” and “GrLivArea” by a wide margin, followed by “TotalBsmtSF”, “2ndFlrSF”, “BsmtFinSF1”, “1stFlrSF”, and garage capacity/area. Neighborhood medians further illustrate clear stratification that aligns with stakeholder expectations and aids communication of price bands.

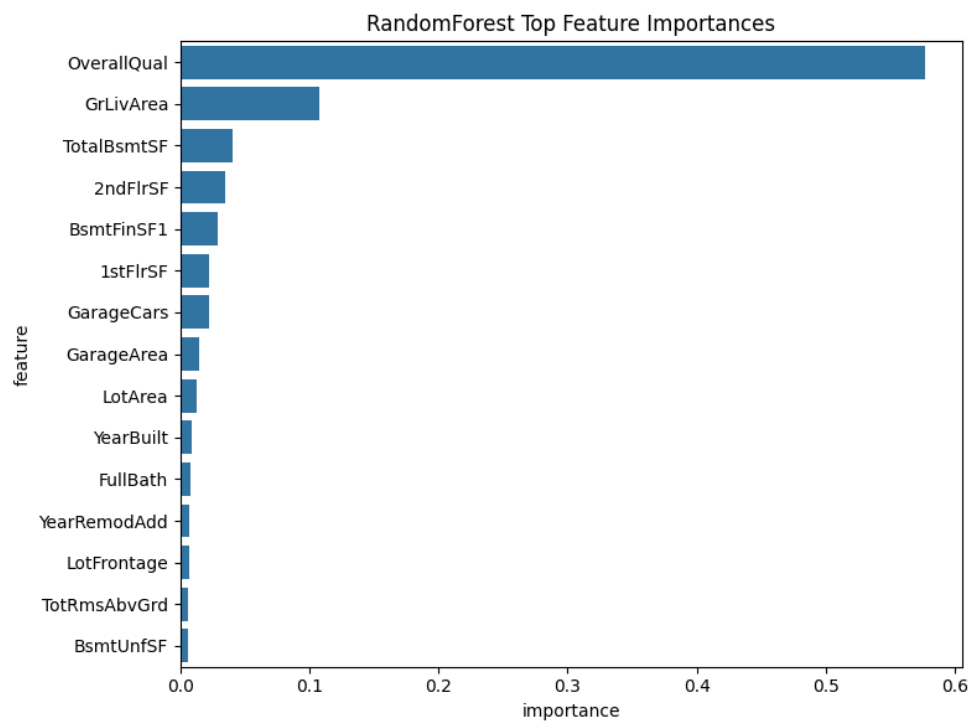


Figure 4. Random Forest top feature importances.

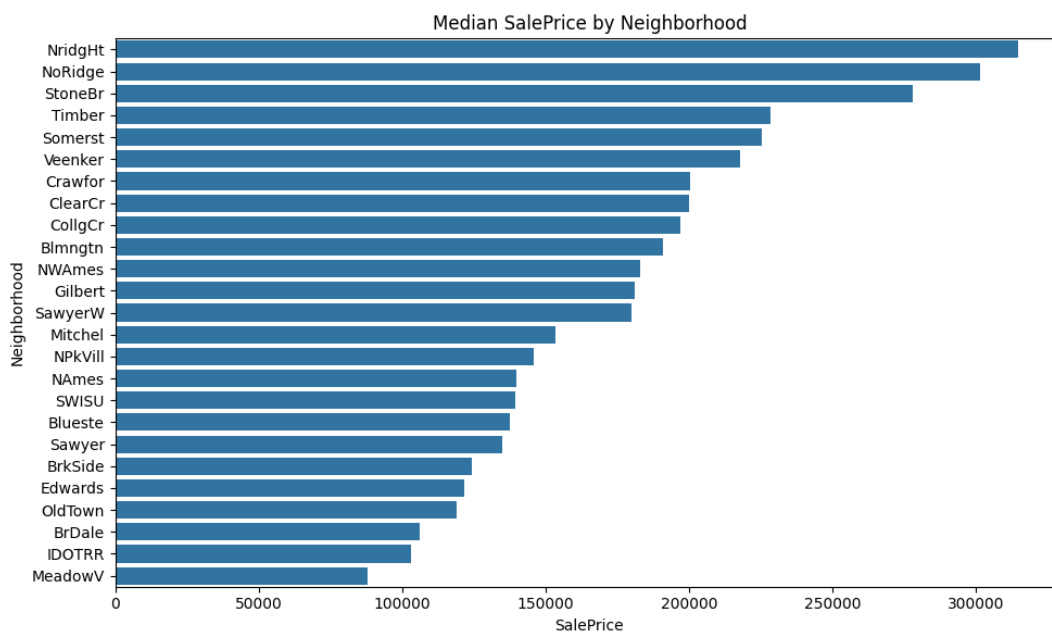


Figure 5. Median sale price by neighborhood.

Final Results

Under 5-fold cross-validation, Gradient Boosting delivers the strongest overall performance (RMSE $\approx$ 28.4K; MAE $\approx$ 16.2K; R2 $\approx$ 0.851), outperforming Random Forest (RMSE $\approx$ 30.3K; MAE $\approx$ 17.8K; R2 $\approx$ 0.839) and the Ridge baseline (RMSE $\approx$ 33.1K; MAE $\approx$ 17.9K; R2 $\approx$ 0.797). A naive log-target Ridge variant under the current preprocessing underperforms on RMSE and R2 (RMSE $\approx$ 53.6K; MAE $\approx$ 17.8K; R2 $\approx$ -0.106). These results support selecting Gradient Boosting for accuracy while retaining Ridge as a transparent baseline; Random Forest remains a competitive non-linear alternative for mixed feature types.

Conclusion

My final evaluation shows that Gradient Boosting achieved the best cross-validated performance, followed by Random Forest and a Ridge baseline. Size and overall quality remain the dominant price drivers, corroborated by correlation structure and tree-based importance. A naïve log-target Ridge variant under the current preprocessing underperformed on RMSE/R2 and would require revised preprocessing and tuning before adoption. For decision support, prefer a tuned Gradient Boosting model for accuracy, retain Ridge for transparency, and keep Random Forest as a robust non-linear comparator. Accompany results with model-agnostic explanations (PDP/ICE/SHAP) to communicate local and global effects.

Assumptions

“NA” often encodes true absence rather than missingness, and the relationships observed are assumed to be relatively stable within the 2006–2010 Ames market represented in the data.

The present draft relies solely on the provided features without external augmentation.

Cross-validation is assumed to reasonably approximate generalization within the dataset’s temporal and geographic scope.

Limitations

The dataset reflects a single city and a limited time window; price dynamics outside this context may differ materially. Multicollinearity among size-related variables complicates coefficient-level interpretation for linear models. One-hot encoding introduces sparsity and the risk of rare categories. The baselines reported here are lightly tuned or not tuned, and the draft does not yet report MAE or R2.

Challenges

The dataset presents several modeling and preprocessing challenges. Categorical variables that use “NA” to indicate absence must be treated with proper encoding to prevent information loss. High-cardinality feature, such as “Neighborhood” require careful, leakage-proof encoding to balance interpretability and model performance. Outliers, distributional skew, and non-linear feature effects necessitate appropriate transformations, as linear models may otherwise underfit. Boosted tree models carry a risk of overfitting without adequate regularization, early stopping, and validation control. Finally, since the Ames data reflect a single geographic area and time frame, model generalizability to other markets or periods will be significantly limited.

Recommendations

For stakeholders, I would emphasize interventions that improve overall quality and habitable area when cost-effective, as these consistently align with higher sale prices. Neighborhood medians should be used as intuitive anchors in pricing conversations, supplemented by model explanations for property-specific adjustments. For production-minded analytics, prefer a tuned gradient boosting model with robust cross-validation, explicit treatment of ordinals, and explainability artifacts in the reporting workflow.

Implementation Plan

- (1) Implement ordinal encodings for quality ranks.
- (2) Develop end-to-end pipelines that model on log-transformed targets where appropriate.
- (3) Conduct systematic hyperparameter search with 5-fold cross-validation.
- (4) Generate SHAP and partial dependence artifacts for both global and local explanations.
- (5) Validate and document the analysis for reproducible reporting.

Ethical Assessment

This project uses de-identified, public market data and provides descriptive, explanatory, and predictive insights intended for educational and exploratory use. The analysis avoids using protected-class attributes and refrains from prescriptive recommendations about individuals. Results should be communicated with uncertainty and limitations, and any deployment should include ongoing monitoring for drift and fairness across relevant cohorts and neighborhoods.

Appendix (Supporting Documentation)

- A supplementary document with anticipated questions and answers from the audience is attached, entitled `DSC_680_Koyrakh_Project2_QA.pdf`.
- The full project code is provided with the submission as a Jupyter Notebook entitled `DSC680_Koyrakh_Project2.ipynb`.
- A presentation of this project for stakeholders is attached:
 - `DSC680_Koyrakh_Project_2_Audio.mp3`
 - `DSC680_Koyrakh_Project_2_Slideshow.pptx`

References

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data Set [Dataset]. Journal of Statistics Education, 19(3). <https://doi.org/10.1080/10691898.2011.11889627>

Kuhn, M. (n.d.). AmesHousing [R package]. Comprehensive R Archive Network (CRAN) and GitHub. Retrieved from <https://github.com/topepo/AmesHousing>

Scikit-learn Developers. (n.d.). Scikit-learn Documentation and Examples for Regression, Model Evaluation, and Pipelines [Software documentation]. Retrieved from <https://scikit-learn.org/stable/>

Kaggle. (n.d.). House Prices - Advanced Regression Techniques [Dataset]. Kaggle. Retrieved from <https://www.kaggle.com/c/house-prices-advanced-regression-techniques>